

claude-windows / 7fce5410-4c57-4ec9-a054-be898bda22da

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7fce5410-4c57-4ec9-a054-be898bda22da\subagents\workflows\wf\_c81dbe7a-7ff\agent-a66a4c5e9735a5315.jsonl`
- SHA-256: `12d306474d7e7fd13534e33e0f482b779b7c5bf921f8f45833cb7318a0f19985`
- Source modified: `2026-06-25T16:57:35+00:00`
- Imported at: `2026-07-05T16:48:26+00:00`
- project: `wf\_c81dbe7a-7ff`
- session_id: `7fce5410-4c57-4ec9-a054-be898bda22da`

Transcript

1. user (2026-06-25T16:57:09.964Z)

Adversarially verify this PR-review finding for PR #380 (branch o1-p11/gemini-remote-file-gc is checked out; read the real code with `git diff origin/master...HEAD` + Read the files). Be skeptical ? default to `isReal=false` / `adjustedSeverity=INVALID` if the finding is wrong, already handled by the code, or a non-issue. Do NOT re-run CI.

Finding [NIT] Inaccurate comment: prod does not use `_safe_delete` in the sweep loop
Location: `tests/test_provider_gemini.py:160`
Detail: The comment 'First delete raises, second succeeds ? `_safe_delete` swallows it.' is wrong. `cleanup_orphaned_files` calls ``self.client.files.delete(name=f.name)`` directly inside its own ``try/except`` (`gemini.py:78-81`); it does not route through ``_safe_delete`` (`gemini.py:37`). The test's assertion (`deleted==1`) is correct, but the rationale misattributes the swallowing mechanism, which could mislead a future maintainer.
Suggested fix: Change the comment to '? the per-file except in `cleanup_orphaned_files` swallows it.'

Return `isReal` + `adjustedSeverity` (`INVALID` if not real) + one-paragraph reasoning grounded in the actual code.

2. assistant (2026-06-25T16:57:12.492Z)

I'll verify this finding by reading the actual code on the PR branch.

3. user (2026-06-25T16:57:13.608Z)

```
backend/main.py | 25 ++++++++
backend/providers/gemini.py | 43 +++++
tests/test_provider_gemini.py | 104 +++++
3 files changed, 172 insertions(+)
```

4. assistant (2026-06-25T16:57:15.454Z)

Let me read the actual code in the relevant files.

5. user (2026-06-25T16:57:16.600Z)

```
diff --git a/backend/providers/gemini.py b/backend/providers/gemini.py
index 2ea0834..428bca8 100644
--- a/backend/providers/gemini.py
+++ b/backend/providers/gemini.py
@@ -2,6 +2,7 @@ import json
import logging
import os
import time
```

```

+from datetime import datetime, timedelta, timezone
+from typing import Tuple

+from google import genai
@@ -40,6 +41,48 @@ class GeminiProvider(AIVideoProvider):
+    except Exception:
+        pass

+    def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
+        """Sweep and delete video files orphaned on the Gemini File API by a previous
+        crashed run (a hard process death between ``files.upload`` and ``_safe_delete``
+        leaves the uploaded clip behind forever ? the per-call cleanup only covers paths
+        that reach a ``finally``/except).

+        Best-effort: lists every file, deletes any whose ``mime_type`` is a video AND whose
+        creation is older than ``grace_sec`` (so we never race a concurrent worker mid-upload).
+        Never raises; returns the count deleted. Intended to run once at startup.

+        Args:
+            grace_sec: only delete files older than this (default 1h). Protects a file a
+                live worker may still be processing.
+        """
+        cutoff = datetime.now(timezone.utc) - timedelta(seconds=grace_sec)
+        deleted = 0
+        try:
+            files = list(self.client.files.list())
+        except Exception as e:
+            logger.warning("[Gemini GC] failed to list remote files (skipping sweep): %s", e)
+            return 0
+        for f in files:
+            try:
+                mime = getattr(f, "mime_type", "") or ""
+                if not mime.startswith("video/"):
+                    continue
+                # f.create_time is an RFC3339 string on the File API.
+                ct = getattr(f, "create_time", None)
+                if ct is None:
+                    # No timestamp ? can't prove it's stale; leave it (safe default).
+                    continue
+                created = datetime.fromisoformat(str(ct).replace("Z", "+00:00"))
+                if created > cutoff:
+                    continue # too recent ? a live worker may own it
+                self.client.files.delete(name=f.name)
+                deleted += 1
+            except Exception as e:
+                logger.debug("[Gemini GC] skipping a file during sweep: %s", e)
+        if deleted:
+            logger.info("[Gemini GC] deleted %d orphaned video file(s) older than %ds.",
+deleted, grace_sec)
+        return deleted

+    def analyze_video(self, video_path: str, prompt: str,
+        chunk_duration: float = 0.0,
+        label: str = "") -> Tuple[list, dict]:

```

6. user (2026-06-25T16:57:17.416Z)

```

1    import json
2    import logging
3    import os
4    import time
5    from datetime import datetime, timedelta, timezone
6    from typing import Tuple
7
8    from google import genai

```

```

9      from google.genai import types
10
11      from backend.infrastructure.execution_policy import (
12          ExecutionPolicy,
13          PolicyTimeout,
14          poll_until,
15          run_bounded,
16      )
17      from backend.providers.base import AIVideoProvider
18
19      logger = logging.getLogger(__name__)
20
21      # The Gemini File API caps uploads at 2GB; stay well under it. Callers must chunk so
22      # each uploaded clip is below this. Oversize uploads are refused (not silently
23      attempted).
24      MAX_UPLOAD_MB = 500
25
26      class GeminiProvider(AIVideoProvider):
27          def __init__(self, api_key: str, model_name: str):
28              super().__init__(api_key, model_name)
29              # Initialize client lazily to avoid connection/auth issues during class
30              instantiation
31              self._client = None
32
33          @property
34          def client(self):
35              if self._client is None:
36                  self._client = genai.Client(api_key=self.api_key)
37              return self._client
38
39          def _safe_delete(self, video_file) -> None:
40              """Best-effort cleanup of an uploaded File (never raises)."""
41              try:
42                  self.client.files.delete(name=video_file.name)
43              except Exception:
44                  pass
45
46          def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
47              """Sweep and delete video files orphaned on the Gemini File API by a previous
48              crashed run (a hard process death between ``files.upload`` and ``_safe_delete``
49              leaves the uploaded clip behind forever ? the per-call cleanup only covers paths
50              that reach a ``finally``/except).
51
52              Best-effort: lists every file, deletes any whose ``mime_type`` is a video AND
53              whose
54              creation is older than ``grace_sec`` (so we never race a concurrent worker mid-
55              upload).
56
57              Never raises; returns the count deleted. Intended to run once at startup.
58
59              Args:
60                  grace_sec: only delete files older than this (default 1h). Protects a file a
61                  live worker may still be processing.
62              """
63              cutoff = datetime.now(timezone.utc) - timedelta(seconds=grace_sec)
64              deleted = 0
65              try:
66                  files = list(self.client.files.list())
67              except Exception as e:
68                  logger.warning("[Gemini GC] failed to list remote files (skipping sweep):
69                  %s", e)
70
71              return 0
72
73          for f in files:
74              try:
75                  mime = getattr(f, "mime_type", "") or ""
76                  if not mime.startswith("video/"):

```

```

69         continue
70         # f.create_time is an RFC3339 string on the File API.
71         ct = getattr(f, "create_time", None)
72         if ct is None:
73             # No timestamp ? can't prove it's stale; leave it (safe default).
74             continue
75         created = datetime.fromisoformat(str(ct).replace("Z", "+00:00"))
76         if created > cutoff:
77             continue # too recent ? a live worker may own it
78         self.client.files.delete(name=f.name)
79         deleted += 1
80     except Exception as e:
81         logger.debug("[Gemini GC] skipping a file during sweep: %s", e)
82     if deleted:
83         logger.info("[Gemini GC] deleted %d orphaned video file(s) older than %ds.",
deleted, grace_sec)
84     return deleted
85
86     def analyze_video(self, video_path: str, prompt: str,
87                       chunk_duration: float = 0.0,
88                       label: str = "") -> Tuple[list, dict]:
89         log_prefix = f"Gemini {label}" if label else "Gemini"
90
91         metrics: dict = {
92             "upload_time": 0.0,
93             "process_time": 0.0,
94             "gen_time": 0.0,
95         }
96
97         # 0. Refuse oversize uploads (Gemini File API limit is 2GB; we cap lower for
safety/speed)
98         try:
99             size_mb = os.path.getsize(video_path) / (1024 * 1024)
100         except OSError:
101             size_mb = 0.0
102         if size_mb > MAX_UPLOAD_MB:
103             logger.error(f"[{log_prefix}] {video_path} is {size_mb:.0f}MB >
{MAX_UPLOAD_MB}MB cap ? ")
104             f"SKIPPED WITHOUT ANALYSIS (no segments for this time range). "
105             f"Re-encode/downscale chunks (indexing.ai_chunk_scale_width) or
chunk smaller.")
106             metrics["error"] = f"oversize: {size_mb:.0f}MB > {MAX_UPLOAD_MB}MB cap"
107             metrics["oversize_skipped"] = True
108             return [], metrics
109
110         # 1. Upload (retry transient network failures, e.g. WinError 10053 / reset
connections)
111         logger.info(f"[{log_prefix}] Uploading {video_path}...")
112         t_upload_start = time.time()
113         video_file = None
114         for attempt in range(3):
115             try:
116                 video_file = self.client.files.upload(file=video_path)
117                 metrics["upload_time"] = time.time() - t_upload_start
118                 logger.info(f"[{log_prefix}] Upload complete in
{metrics['upload_time']:.1f}s. URI: {video_file.uri}")
119                 break
120             except Exception as e:
121                 logger.warning(f"[{log_prefix}] Upload attempt {attempt + 1}/3 failed:
{e}")
122                 if attempt < 2:
123                     time.sleep(2 * (attempt + 1))
124         if video_file is None:
125             logger.error(f"[{log_prefix}] Upload failed after 3 attempts.")
126             metrics["error"] = "upload failed after 3 attempts"

```

```

127         return [], metrics
128
129     # 2. Wait for processing on Google servers ? a BOUNDED poll (a stuck file must
not hang
130     # the worker forever). Per-poll logs at DEBUG (no INFO spam); see
EXECUTION_POLICY.md.
131     t_process_start = time.time()
132     process_policy = ExecutionPolicy(poll_every_n_sec=10.0, max_wait_sec=1200.0) #
20 min cap
133
134     def _processed():
135         nonlocal video_file
136         video_file = self.client.files.get(name=video_file.name)
137         return None if video_file.state.name == "PROCESSING" else video_file
138
139     try:
140         video_file = poll_until(_processed, process_policy,
141                                label=f"{log_prefix} processing", logger=logger)
142     except PolicyTimeout:
143         logger.error(f"[{log_prefix}] Processing exceeded
{process_policy.max_wait_sec:.0f}s ? giving up.")
144         self._safe_delete(video_file)
145         metrics["error"] = f"processing timed out after
{process_policy.max_wait_sec:.0f}s"
146         return [], metrics
147     except Exception as e:
148         logger.error(f"[{log_prefix}] Status check failed: {e}")
149         self._safe_delete(video_file)
150         metrics["error"] = f"status check failed: {e}"
151         return [], metrics
152     if video_file.state.name == "FAILED":
153         logger.error(f"[{log_prefix}] Video processing failed:
{video_file.error.message}")
154         self._safe_delete(video_file)
155         metrics["error"] = f"video processing failed: {video_file.error.message}"
156         return [], metrics
157
158     metrics["process_time"] = time.time() - t_process_start
159     logger.info(f"[{log_prefix}] Processing finished in
{metrics['process_time']:.1f}s.")
160
161     # 3. Call Gemini under a HARD per-attempt timeout (op_timeout_sec ? the hang-
killer that was
162     # missing: a generate() that never returns previously blocked forever) + retry
the 503/network
163     # FAILURE waves (capacity 503s on the flagship flash come in bursts; ~6
attempts).
164     logger.info(f"[{log_prefix}] Submitting multimodal request to
{self.model_name}...")
165     t_gen_start = time.time()
166     gen_policy = ExecutionPolicy(op_timeout_sec=180.0, max_retries=5,
backoff_base_sec=3.0)
167     raw_text = None
168
169     def _generate():
170         return self.client.models.generate_content(
171             model=self.model_name,
172             contents=[video_file, prompt],
173             config=types.GenerateContentConfig(
174                 response_mime_type="application/json", temperature=0.0),
175         )
176
177     try:
178         response = run_bounded(_generate, gen_policy, label=f"{log_prefix}
generate", logger=logger)

```

```

179         raw_text = response.text
180         metrics["gen_time"] = time.time() - t_gen_start
181         logger.info(f"[{log_prefix}] Response received in
{metrics['gen_time']:.1f}s:\n{raw_text}")
182     except Exception as e:                                # PolicyTimeout (hang) or the last 503 ?
both end here
183         logger.error(f"[{log_prefix}] generate gave up after retries: {e}")
184         metrics["error"] = f"generate failed after retries: {e}"
185         metrics["generate_failed"] = True
186         self._safe_delete(video_file)
187
188     if not raw_text:
189         return [], metrics
190
191     # 4. Parse JSON
192     try:
193         segments = json.loads(raw_text)
194         if not isinstance(segments, list):
195             logger.error(f"[{log_prefix}] Response is not a JSON array.")
196             metrics["error"] = "response is not a JSON array"
197             return [], metrics
198         return segments, metrics
199     except Exception as e:
200         logger.error(f"[{log_prefix}] JSON parse failed: {e}")
201         metrics["error"] = f"JSON parse failed: {e}"
202         return [], metrics
203

```

7. user (2026-06-25T16:57:17.667Z)

```

1     import logging
2     from unittest.mock import MagicMock, patch
3
4     from backend.providers.gemini import GeminiProvider
5
6
7     def test_gemini_provider_generate_failure_is_flagged():
8         """A generate() give-up (per-attempt timeout / 503 burst after retries) must set
9         metrics['generate_failed'] so a FAILED chunk isn't silently mistaken for 'no
rallies'
10         (the silent-partial-result guard the segmenter surfaces loudly)."""
11         with patch("backend.providers.gemini.genai") as mock_genai, \
12             patch("backend.providers.gemini.run_bounded", side_effect=RuntimeError("503
exhausted")):
13             client_mock = MagicMock()
14             mock_genai.Client.return_value = client_mock
15             file_mock = MagicMock()
16             file_mock.state.name = "ACTIVE"
17             file_mock.name = "dummy_file"
18             client_mock.files.upload.return_value = file_mock
19             client_mock.files.get.return_value = file_mock
20
21             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
22             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt",
chunk_duration=10.0)
23
24             assert segments == []                                # no analysis for this chunk
25             assert metrics.get("generate_failed") is True        # ...but the failure is FLAGGED,
not silent
26
27
28     def test_gemini_provider_success():
29         # Mock GenAI client
30         with patch("backend.providers.gemini.genai") as mock_genai:

```

```

31         client_mock = MagicMock()
32         mock_genai.Client.return_value = client_mock
33
34         # Mock file upload
35         file_mock = MagicMock()
36         file_mock.uri = "gs://dummy"
37         file_mock.name = "dummy_file"
38         file_mock.state.name = "ACTIVE"
39
40         client_mock.files.upload.return_value = file_mock
41         client_mock.files.get.return_value = file_mock
42
43         # Mock generate_content response
44         response_mock = MagicMock()
45         response_mock.text = '[{"start_time": 1.0, "end_time": 2.0}]'
46         client_mock.models.generate_content.return_value = response_mock
47
48         provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
49         segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt",
chunk_duration=10.0)
50
51         assert len(segments) == 1
52         assert segments[0]["start_time"] == 1.0
53         assert metrics["upload_time"] >= 0
54         assert metrics["process_time"] >= 0
55         assert metrics["gen_time"] >= 0
56
57         # Ensure delete is called
58         client_mock.files.delete.assert_called_with(name="dummy_file")
59
60     def test_gemini_provider_invalid_json():
61         with patch("backend.providers.gemini.genai") as mock_genai:
62             client_mock = MagicMock()
63             mock_genai.Client.return_value = client_mock
64             file_mock = MagicMock()
65             file_mock.state.name = "ACTIVE"
66             client_mock.files.upload.return_value = file_mock
67             client_mock.files.get.return_value = file_mock
68
69             response_mock = MagicMock()
70             response_mock.text = '{"not_a_list": true}'
71             client_mock.models.generate_content.return_value = response_mock
72
73             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
74             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")
75             assert len(segments) == 0
76
77     def test_gemini_provider_rejects_oversize_before_upload(monkeypatch, caplog):
78         with patch("backend.providers.gemini.genai") as mock_genai:
79             client_mock = MagicMock()
80             mock_genai.Client.return_value = client_mock
81             # 600MB > MAX_UPLOAD_MB (500) -> refuse without ever uploading
82             monkeypatch.setattr("backend.providers.gemini.os.path.getsize",
lambda p: 600 * 1024 * 1024)
83             provider = GeminiProvider(api_key="k", model_name="gemini-flash-latest")
84             with caplog.at_level(logging.ERROR, logger="backend.providers.gemini"):
85                 segments, metrics = provider.analyze_video("big.mp4", "p")
86             assert segments == []
87             client_mock.files.upload.assert_not_called()
88             # The refusal must be LOUD and machine-visible: callers aggregate the metrics
89             # flag into the per-video report (oversize_clip_skipped rule).
90             assert metrics["oversize_skipped"] is True
91             assert any("SKIPPED WITHOUT ANALYSIS" in r.message for r in caplog.records)

```

```

93
94
95     def test_gemini_provider_upload_fail():
96         with patch("backend.providers.gemini.genai") as mock_genai:
97             client_mock = MagicMock()
98             mock_genai.Client.return_value = client_mock
99             client_mock.files.upload.side_effect = Exception("Upload err")
100
101             provider = GeminiProvider(api_key="dummy_api_key",
model_name="gemini-2.5-flash")
102             segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")
103             assert len(segments) == 0
104
105
106
107     # --- 01/P11: Gemini remote-file GC (startup orphan sweep) ---
108
109     def _file_mock(name, mime, create_time_iso, state="ACTIVE"):
110         f = MagicMock()
111         f.name = name
112         f.mime_type = mime
113         f.create_time = create_time_iso
114         f.state = MagicMock()
115         f.state.name = state
116         return f
117
118
119     def test_cleanup_orphaned_files_deletes_stale_videos_only():
120         """Deletes video files older than the grace window; leaves non-videos and fresh
files."""
121         from datetime import datetime, timedelta, timezone
122         stale = (datetime.now(timezone.utc) - timedelta(hours=2)).isoformat()
123         fresh = (datetime.now(timezone.utc) - timedelta(minutes=5)).isoformat()
124         files = [
125             _file_mock("old-vid", "video/mp4", stale),          # stale video -> deleted
126             _file_mock("fresh-vid", "video/quicktime", fresh), # fresh video -> kept (grace)
127             _file_mock("old-img", "image/png", stale),          # non-video -> kept
128             _file_mock("notts", "video/mp4", None),             # no timestamp -> kept (safe)
129         ]
130         with patch("backend.providers.gemini.genai"):
131             provider = GeminiProvider(api_key="k", model_name="m")
132             provider._client = MagicMock()
133             provider._client.files.list.return_value = iter(files)
134             deleted = provider.cleanup_orphaned_files(grace_sec=3600)
135             assert deleted == 1
136             provider._client.files.delete.assert_called_once_with(name="old-vid")
137
138
139     def test_cleanup_orphaned_files_list_failure_is_swallowed():
140         """A list() failure must not raise ? the sweep is best-effort and must not block
startup."""
141         with patch("backend.providers.gemini.genai"):
142             provider = GeminiProvider(api_key="k", model_name="m")
143             provider._client = MagicMock()
144             provider._client.files.list.side_effect = RuntimeError("network")
145             assert provider.cleanup_orphaned_files() == 0
146
147
148     def test_cleanup_orphaned_files_per_file_delete_failure_is_swallowed():
149         """A delete() failure on one file must not abort the sweep."""
150         from datetime import datetime, timedelta, timezone
151         stale = (datetime.now(timezone.utc) - timedelta(hours=2)).isoformat()
152         files = [
153             _file_mock("a", "video/mp4", stale),
154             _file_mock("b", "video/mp4", stale),

```



```

155     ]
156     with patch("backend.providers.gemini.genai"):
157         provider = GeminiProvider(api_key="k", model_name="m")
158         provider._client = MagicMock()
159         provider._client.files.list.return_value = iter(files)
160         # First delete raises, second succeeds ? _safe_delete swallows it.
161         provider._client.files.delete.side_effect = [RuntimeError("boom"), None]
162         deleted = provider.cleanup_orphaned_files(grace_sec=3600)
163     assert deleted == 1 # only the second file counted
164
165
166
167     # --- 01/P11: main.py startup wiring (_maybe_sweep_gemini_orphans) ---
168
169     def test_maybe_sweep_skips_non_gemini_provider(monkeypatch):
170         """A non-Gemini provider must short-circuit (no client built, no sweep)."""
171         import backend.main as main_mod
172         cfg = MagicMock()
173         cfg.ai_provider = "openai"
174         called = []
175         monkeypatch.setattr(main_mod.provider_registry, "get",
176                             lambda *a, **k: called.append(k) or MagicMock())
177         monkeypatch.setenv("GEMINI_API_KEY", "k")
178         main_mod._maybe_sweep_gemini_orphans(cfg)
179         assert called == [] # provider never instantiated
180
181
182     def test_maybe_sweep_skips_when_no_api_key(monkeypatch):
183         """No GEMINI_API_KEY -> no sweep (can't auth)."""
184         import backend.main as main_mod
185         cfg = MagicMock()
186         cfg.ai_provider = "gemini"
187         monkeypatch.delenv("GEMINI_API_KEY", raising=False)
188         called = []
189         monkeypatch.setattr(main_mod.provider_registry, "get",
190                             lambda *a, **k: called.append(k) or MagicMock())
191         main_mod._maybe_sweep_gemini_orphans(cfg)
192         assert called == []
193
194
195     def test_maybe_sweep_runs_and_swallows_errors(monkeypatch):
196         """Gemini + key present -> sweep runs; a provider error is swallowed (never blocks
197 startup)."""
198         import backend.main as main_mod
199         cfg = MagicMock()
200         cfg.ai_provider = "gemini"
201         cfg.ai_model = "gemini-2.5-flash"
202         provider = MagicMock()
203         provider.cleanup_orphaned_files.side_effect = RuntimeError("api down")
204         monkeypatch.setattr(main_mod.provider_registry, "get", lambda *a, **k: provider)
205         monkeypatch.setenv("GEMINI_API_KEY", "k")
206         # Must not raise.
207         main_mod._maybe_sweep_gemini_orphans(cfg)
208         provider.cleanup_orphaned_files.assert_called_once()

```

8. assistant (2026-06-25T16:57:27.499Z)

I have all the information I need. Let me verify the finding precisely.

The finding is about line 160 of `tests/test\_provider\_gemini.py`:

...

First delete raises, second succeeds ? _safe_delete swallows it.

...

Looking at the actual code:

1. `cleanup_orphaned_files` (gemini.py:44-84)**: Inside its sweep loop at line 78, it calls `self.client.files.delete(name=f.name)` **directly**, NOT through `_safe_delete`. The exception that a failing delete throws is caught by the per-file `try/except` at lines 66-81 (`except Exception as e: logger.debug(...)` at line 80).

2. `_safe_delete` (gemini.py:37-42)**: A separate method that is only called in `analyze_video` (lines 144, 149, 154, 186), never in `cleanup_orphaned_files`.

3. **The test** (`test_cleanup_orphaned_files_per_file_delete_failure_is_swallowed`, lines 148-163)**: It exercises `cleanup_orphaned_files`, where the first `delete` raises `RuntimeError("boom")` and the second succeeds. The swallowing mechanism that makes `deleted == 1` true is the **per-file** `except Exception` block inside the sweep loop** (line 80), NOT `_safe_delete`.

The finding is factually correct: the comment misattributes the swallowing to `_safe_delete`, when in reality `cleanup_orphaned_files` never routes through `_safe_delete` ? it catches the exception in its own inline `try/except`. This is a genuine inaccuracy in a test comment that could mislead a future maintainer into thinking the two code paths share the same cleanup helper. The assertion itself is correct. Severity NIT is appropriate ? it's a comment-only inaccuracy in a test, with no functional impact.

9. user (2026-06-25T16:57:31.413Z)

Structured output provided successfully

10. assistant (2026-06-25T16:57:35.185Z)

Verified. The finding is real and correctly classified as NIT.

The test comment at `tests/test_provider_gemini.py:160` misattributes the exception-swallowing to `_safe_delete`, but `cleanup_orphaned_files` (`backend/providers/gemini.py:44-84`) calls `self.client.files.delete()` directly at line 78 inside its own per-file `try/except` (the `except` at line 80 is what swallows the error). `_safe_delete` (lines 37-42) is only used in `analyze_video`. The assertion is correct; only the comment's rationale is wrong, so it's a harmless cosmetic inaccuracy.